Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 24 GB of memory. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the accuracy during training.

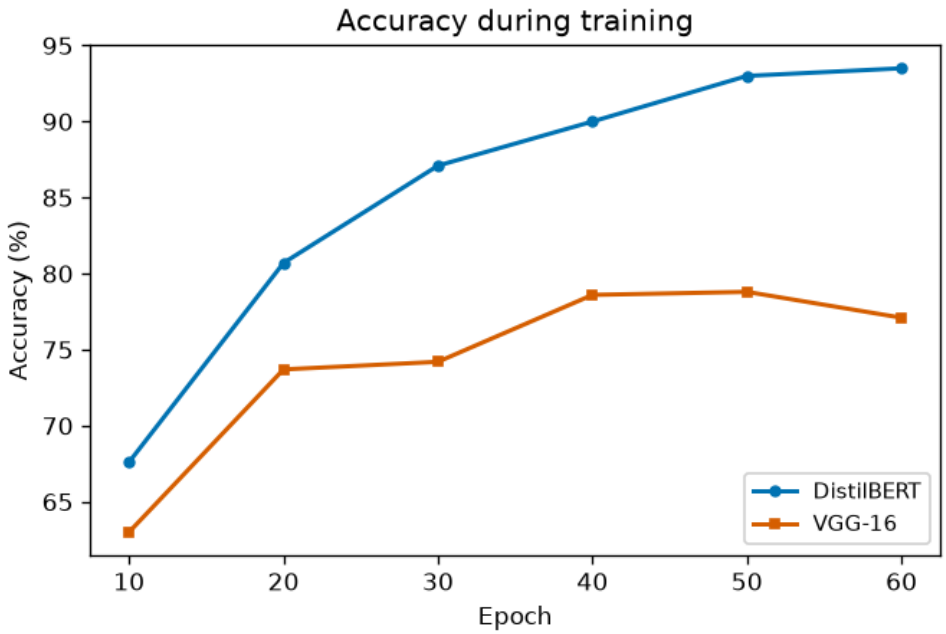


Figure 1: Accuracy during training for the models in this experiment.

DistilBERT outperforms VGG-16 by 5 percentage points at epoch 10. As shown in Figure 1, at epoch 30, DistilBERT is 12.9 points better than VGG-16. At epoch 10, DistilBERT achieves about 70% accuracy. VGG-16 reaches 78.8% accuracy at epoch 50 (Figure 1).

Figure 2 shows the training loss.

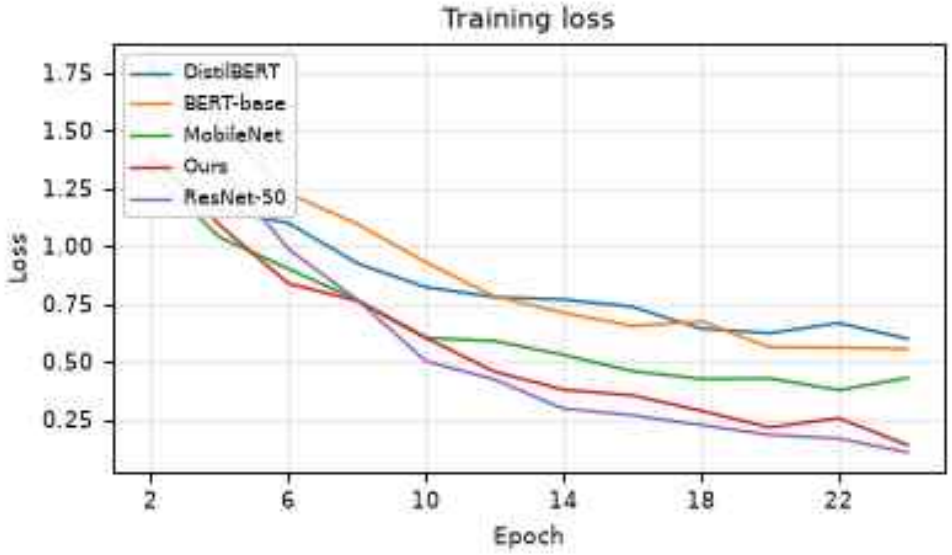


Figure 2: Training loss for the models in this experiment.

MobileNet has a loss of 0.377 at epoch 14. As shown in Figure 2, at epoch 22, the loss of DistilBERT is 1.036. At epoch 10, the loss of BERT-base is 0.602 (Figure 2).

3 Discussion

On SST-2, XGBoost obtains an accuracy of 73.6%. Overall, the results are in line with what we expected from the literature.